Simulating bosonic interferences on a superconducting quantum computer

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Abstract

In this article we attempt to simulate bosonic interference properties with the use of qubit-based superconducting quantum computer. The studies are performed for the truncation of the Fock space involving the $|0\rangle, |1\rangle, |2\rangle$ and $|3\rangle$ states. A general quantum beam splitter is then represented as a quantum circuit involving 4 qubits. The quantum simulations perforemed allowed to observe the Hong–Ou–Mandel effect. Furthermore, analysis beyond the symmetric case has been performed.... The simulations have been performed on the VLQ superconducting quantum computer.

1 Introduction

2 Quantum beam splitter and MZI

3 Truncated Fock space

In the mode representation, the truncated Fock space, can be spanned by the following basis vectors:

$$|0\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \quad |1\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}, \quad |2\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad |3\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}. \tag{1}$$

The truncated creation and annihilation operators for read:

$$\hat{b} = \begin{pmatrix}0 & 1 & 0 & 0\\0 & 0 & \sqrt{2} & 0\\0 & 0 & 0 & \sqrt{3}\\0 & 0 & 0 & 0\end{pmatrix}, \quad \hat{b}^\dagger = \begin{pmatrix}0 & 0 & 0 & 0\\1 & 0 & 0 & 0\\0 & \sqrt{2} & 0 & 0\\0 & 0 & \sqrt{3} & 0\end{pmatrix}. \tag{2}$$

The operators are adjoint to each other and satisfy the following algebra:

$$[\hat{b}, \hat{b}^\dagger] = \begin{pmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & 1 & 0\\0 & 0 & 0 & -3\end{pmatrix}. \tag{3}$$

The occupation number operator is:

$$\hat{N} := \hat{b}^\dagger \hat{b} = \begin{pmatrix}0 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & 2 & 0\\0 & 0 & 0 & 3\end{pmatrix}. \tag{4}$$

4 Truncated quantum beam splitter

A beam splitter (see Fig. 1) is characterized by two input channels (denoted as $1, 2$) and two output channels (denoted as $3, 4$). Associated with these channels are the Fock spaces:

$$\mathcal{H}_i = \text{span}\{|0\rangle_i, |1\rangle_i, |2\rangle_i, |3\rangle_i\}, \tag{5}$$

which describe the states of light. In these spaces, we can define the creation operators $\hat{a}_i^\dagger$ and annihilation operators $\hat{b}_i$, for $i = 1, 2, 3, 4$, satisfying the standard commutation relations:

$$[\hat{b}_i, \hat{b}_j^\dagger] = (\hat{\mathbb{I}} - 4\,|3\rangle\langle 3|)\,\delta_{ij}, \tag{6}$$

$$[\hat{b}_i, \hat{b}_j] = 0 = [\hat{b}_i^\dagger, \hat{b}_j^\dagger]. \tag{7}$$

Moreover, the number operators are defined as:

$$\hat{N}_i := \hat{b}_i^\dagger \hat{b}_i.$$

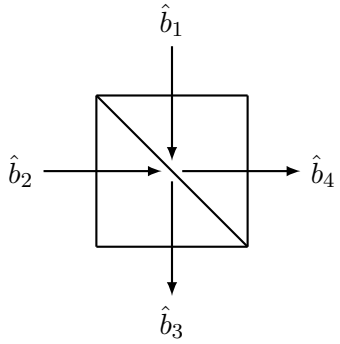


Figure 1: Schematic diagram of the quantum beam splitter.

The transformation operator for the symmetric beam splitter can now be expressed as:

$$\hat{U}_{BS} = \exp\left[\theta\left(e^{i\phi}\hat{b}_1^\dagger \otimes \hat{b}_2 - e^{-i\phi}\hat{b}_1 \otimes \hat{b}_2^\dagger\right)\right], \tag{8}$$

where the phase ϕ dependence can be extracted as follows:

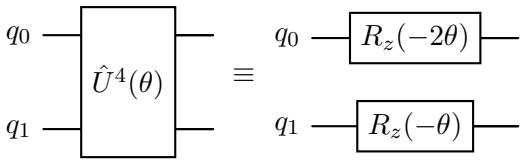
$$\hat{U}_{BS}(\theta, \phi) = (\hat{U}(\phi)^\dagger \otimes \hat{\mathbb{I}})\hat{U}_{BS}(\theta, 0)(\hat{U}(\phi) \otimes \hat{\mathbb{I}}). \tag{9}$$

Here $\hat{U}$ crresponds to the phase shift operator:

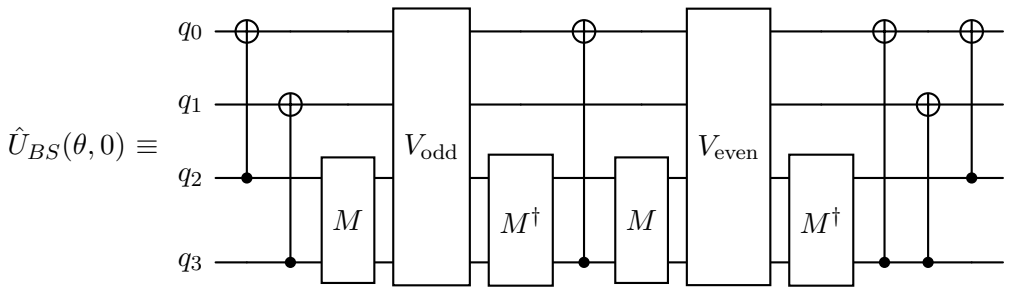
$$\hat{U}(\phi) = \exp(-i\phi\hat{N}) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{-i\phi} & 0 & 0 \\ 0 & 0 & e^{-i2\phi} & 0 \\ 0 & 0 & 0 & e^{-i3\phi} \end{pmatrix}. \tag{10}$$

$$\tag{11}$$

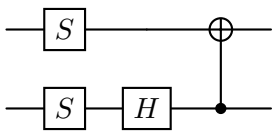
So that, the quantum circuit is:



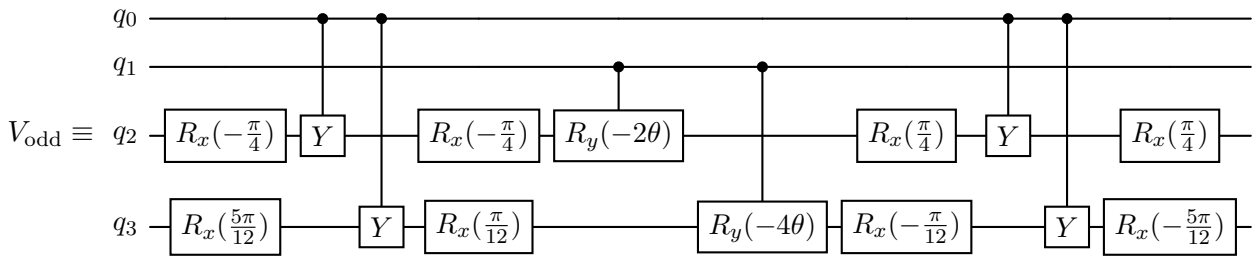
The optimized circuit for the beam splitter takes the following form:



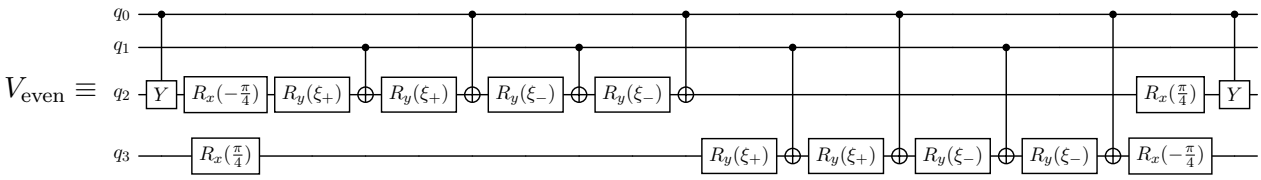
where:

$M \equiv$ 

Here, V_{odd} implements U_1, U_3 and U_5 sectors and acts only for $q_1 = 1$:

$V_{\text{odd}} \equiv$ 

Similarly, V_{even} implements U_2 and U_4 sectors and acts only for $q_1 = 0$:

$V_{\text{even}} \equiv$ 

where:

$$\xi_{\pm} = -\frac{\sqrt{3} \pm 1}{2}\,\theta. \tag{12}$$

4.1 Quantum circuit for $|1\rangle\,|1\rangle$ input state

5 Experimental setup

Description of the VLQ architecture and parameters

6 Quantum simulations of the bosonic interferences

Let us assume that the initial photonic state is $|1\rangle\,|1\rangle$

The circuits simplifies for this particular input. Only the even sector contribuite.

Then, perform, quantum simulations of the quantum circuit $\hat{U}_{BS}(\theta, 0)$ for $\theta \in [0, \pi/2]$. Here, $\theta = \pi/4$ corresponds to the symmetric beam splitter. Specifically, let us consider $\theta = \frac{\pi}{32}n$, with $n = 0, 1, ..., 16$.

Analyse probabilies of the photonic output states $|2\rangle\,|0\rangle$, $|1\rangle\,|1\rangle$, $|0\rangle\,|2\rangle$, as a function of θ. Plot it as a function of θ and compare with the theoretical prediction:

$$P(2,0) = \frac{1}{2}\sin^2(2\theta), \tag{13}$$

$$P(1,1) = \cos^2(2\theta), \tag{14}$$

$$P(0,2) = \frac{1}{2}\sin^2(2\theta), \tag{15}$$

so that $P(2,0) + P(1,1) + P(0,1) = \sin^2(2\theta) + \cos^2(2\theta) = 1$.

7 Summary

Acknowledgements

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Appendix: Definitions of the quantum gates

- X, Y, Z

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{16}$$

- H

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{17}$$

- S

$$S = R_z\left(\frac{\pi}{2}\right) = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \tag{18}$$

- R_x

$$R_x(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -i\,\sin\left(\frac{\theta}{2}\right) \\ -i\,\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix} \tag{19}$$

- R_y

$$R_y(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix} \tag{20}$$

- R_z

$$R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \tag{21}$$

- U_1

$$U_1(\phi) = e^{i\phi/2} R_z(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \tag{22}$$

- U_2

$$U_2(\phi, \delta) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -\exp(i\delta) \\ \exp(i\phi) & \exp(i(\phi+\delta)) \end{pmatrix} \tag{23}$$

- U_3

$$U_3(\theta, \phi, \delta) = \begin{pmatrix} \cos(\theta/2) & -\exp(i\delta)\sin(\theta/2) \\ \exp(i\phi)\sin(\theta/2) & \exp(i(\phi+\delta))\cos(\theta/2) \end{pmatrix} \tag{24}$$

- $CNOT$

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{25}$$

- $SWAP$

$$SWAP = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{26}$$

- G

$$G(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta/2) & -\sin(\theta/2) & 0 \\ 0 & \sin(\theta/2) & \cos(\theta/2) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{27}$$

References